Glyph Ascent - Technical Sheet

Tech used

One self-contained HTML file (about 29 KB), no libraries or external assets. DOM cards with CSS transforms for crisp, large touch targets; HTML5 Canvas 2D for an animated rune-field background with additive particle bursts and screenshake, and for procedurally drawn, animated monsters (glow, breathing, blinking eyes). Web Audio API for every sound: synthesized effects and a procedural minor-key track that speeds up and adds hats in elite and boss fights. localStorage for meta progress and mute. Portrait-first layout, safe-area aware, haptics where supported.

Systems

Pure, seeded game core (GA module) separated from the UI so it can be simulated headless. 18 cards (attacks, skills, powers) with damage, multi-hit, area, block, draw, energy, heal, strength, thorns, poison, weak and vulnerable. 7 foe types (4 normal, 2 elites, 1 boss) with cycling telegraphed intents and floor scaling. 12-floor branching path with battles, elites, rests and shops; card rewards, gold, card removal; 4-track permanent upgrade forge.

Audit 1: Balance (simulated play)

A greedy bot played 300 runs at three upgrade levels. First version: 0% wins even fully upgraded, average death on floor 7-10. Fixes: gentler enemy HP and damage scaling, boss HP 190 to 130 and its big hit 28 to 20, normal foes hit 15-20% softer, 3-foe packs only from floor 6, +6 heal after each win, rest heal 30% to 35%, start HP 70 to 80. Result: fresh 4% bot wins (average floor 10.6), mid upgrades 19%, full upgrades 53%, about 65-100 embers per run. Human players who read intents should do noticeably better than the bot.

Audit 2: Bugs

A headless harness with a mock DOM played 40 full runs through battles, elites, rests, card removal, shops, rewards, skips, deaths and the menu: 0 errors, no stuck fights. Guards added so a win can never pay out twice (the last card and End Turn could both trigger it), End Turn is locked while enemy attacks animate, a dead target automatically switches to a living foe, and audio is optional so browsers without Web Audio do not crash.

Audit 3: Player experience

Every foe shows its next move with exact damage after Weak/Vulnerable, so losses are never a surprise. Added a first-fight tap tutorial, a low-health warning, floor and boss banners, floating damage and block numbers, red hit flash and vibration, greyed-out unaffordable cards with a Not enough energy callout, a preview of the next floor on the path screen, a deck viewer, a mute button, and audio that pauses when the app is backgrounded.

Testing note

Tested with simulated play and a headless harness, not yet on a physical phone.